INDIA’S AGRICULTURAL CROP PRODUCTION ANALYSIS (1997-2021)

Submitted by:

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Submitted To:

Smart Bridge Internship Program

1. INTRODUCTION

1.1 Project Overview

Agriculture is one of the most important sectors of the Indian economy, providing employment to a large portion of the population and contributing significantly to the country's GDP. Understanding agricultural production trends helps farmers, researchers, policymakers, and businesses make informed decisions related to crop planning, food security, and resource management.

This project, **India's Agricultural Crop Production Analysis (1997–2021)**, uses Tableau to analyze historical agricultural data collected across different states and years. The dataset includes information about crop production, cultivated area, crop types, seasons, and production quantities.

Using Tableau, interactive dashboards and stories are created to visualize crop production patterns, compare state-wise and year-wise performance, identify the highest-producing crops, and analyze seasonal trends. These visualizations help users understand complex agricultural data quickly and support data-driven decision-making.

The project demonstrates how business intelligence tools such as Tableau can transform raw agricultural data into meaningful insights through charts, maps, filters, and dashboards.

1.2 Purpose

The main purpose of this project is to analyze India's agricultural crop production data using Tableau and present the results through interactive visualizations. The project aims to identify production trends, compare crop performance across different states and years, and provide valuable insights that support better agricultural planning and decision-making.

The objectives of this project are:

- To analyze agricultural crop production data from 1997 to 2021.
- To compare crop production across different Indian states.
- To identify the top-performing crops and regions.
- To study year-wise and season-wise production trends.
- To create interactive dashboards and stories using Tableau.

- To help users explore agricultural data easily through filters and visualizations.
- To support data-driven decisions for farmers, researchers, and policymakers.

Overall, this project demonstrates how Tableau can be used to convert raw agricultural data into clear, meaningful, and interactive reports that improve understanding of India's agricultural production.

2. IDEATION PHASE

2.1 Problem Statement

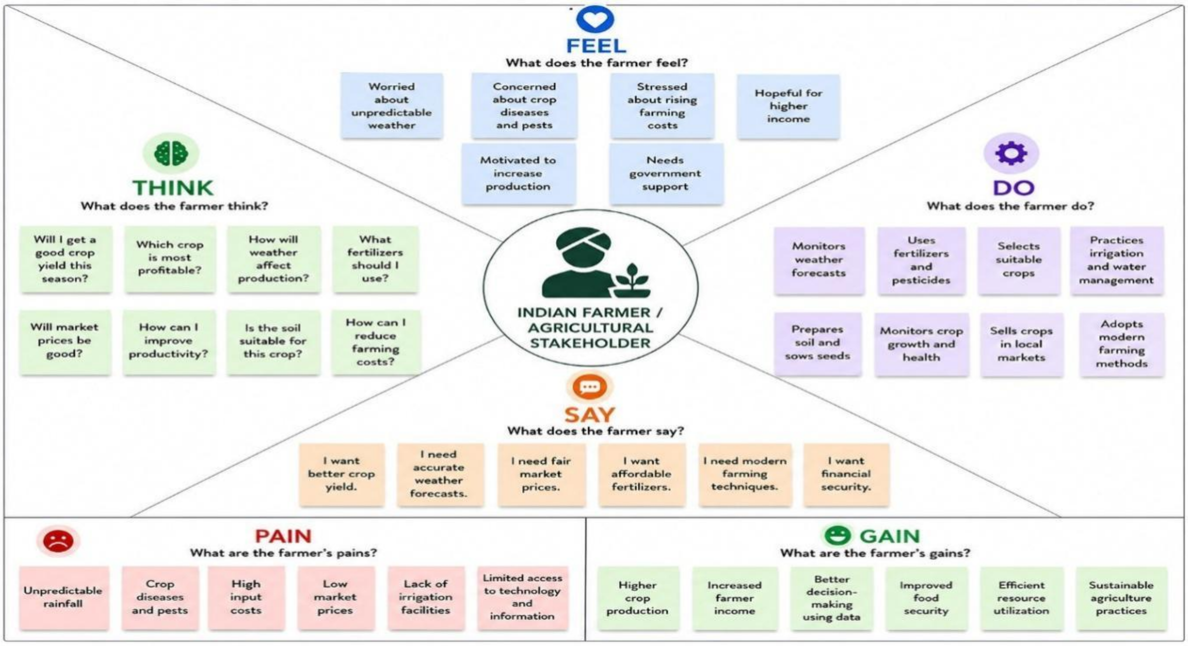
India is one of the world's largest agricultural producers, generating vast amounts of crop production data every year. However, this data is often stored in large tables and spreadsheets, making it difficult to identify trends, compare state-wise production, or analyze seasonal and yearly variations.

The lack of effective data visualization limits the ability of farmers, researchers, policymakers, and agricultural organizations to make informed decisions. Without proper analysis, it becomes challenging to identify high-performing crops, monitor production changes, and plan agricultural strategies.

This project addresses these challenges by using Tableau to transform raw agricultural data into interactive dashboards and stories. The visualizations enable users to explore crop production patterns, compare regions and years, and gain meaningful insights for better agricultural planning and decision-making.

2.2 Empathy Map Canvas

The Empathy Map helps understand the needs and expectations of users who analyze agricultural data.



2.3 Brainstorming

During the brainstorming phase, several ideas were discussed to determine the best way to analyze India's agricultural crop production data. The focus was on creating an interactive and user-friendly Tableau dashboard that provides meaningful insights.

Brainstorming Ideas

- Analyze crop production from 1997 to 2021.
- Compare agricultural production across different states.
- Identify the top-producing crops in India.
- Study year-wise production trends.
- Analyze seasonal crop production (Kharif, Rabi, etc.).
- Compare cultivated area and production.
- Use maps to visualize state-wise crop production.
- Create interactive filters for state, crop, and year selection.
- Build dashboards and stories for easy interpretation of data.

- Generate insights that support agricultural planning and policy decisions.

3. REQUIREMENT ANALYSIS

3.1 Project Workflow (Customer Journey Map)

The following table represents the workflow followed in the development of the India's Agricultural Crop Production Analysis (1997–2021) project using Tableau.

Phase	Activity	Output
Data Collection	Import the agricultural crop production dataset (1997–2021) into Tableau.	Dataset loaded successfully.
Data Understanding	Review the dataset, identify columns, data types, and missing values.	Better understanding of the data structure.
Data Cleaning	Remove duplicate records, handle null values, correct data types, and prepare the dataset for analysis.	Clean and reliable dataset.
Data	Create charts, graphs, maps, and other	Interactive worksheets.

Phase	Activity	Output
Visualization	visualizations to analyze crop production trends.	
Dashboard Creation	Combine multiple visualizations into an interactive dashboard with filters.	User-friendly dashboard for analysis.
Story Creation	Arrange dashboards into a Tableau Story to present insights step by step.	Complete project story with key findings.
Result Analysis	Interpret the visualizations and summarize the major findings.	Meaningful insights and conclusions.

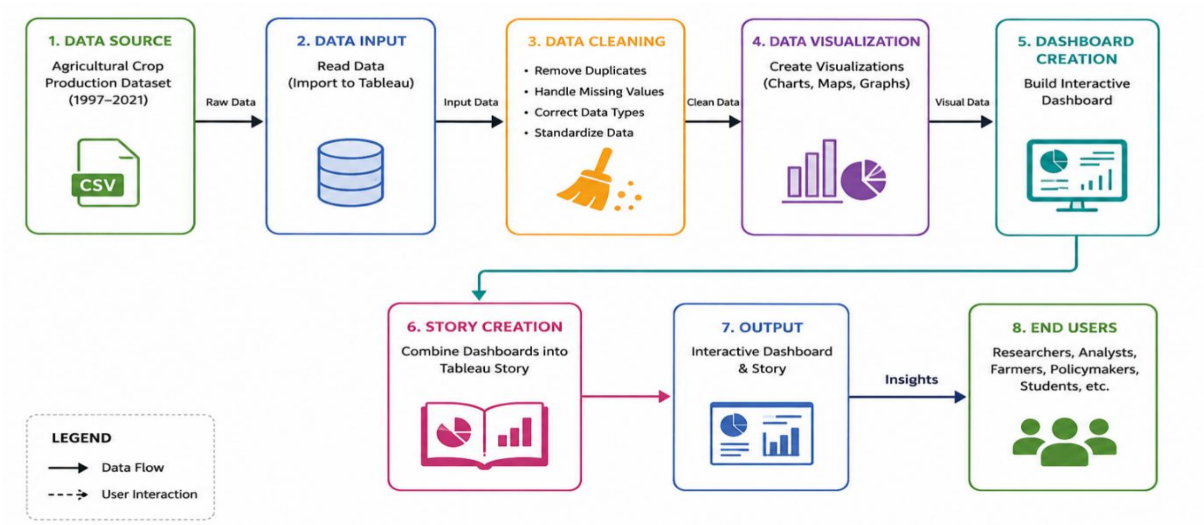
3.2 Solution Requirement

Functional Requirements vs Non-Functional Requirements

Functional Requirements	**Non-Functional Requirements**
Import the agricultural crop production dataset.	Easy-to-use and interactive user interface.
Clean and prepare the data for analysis.	Fast dashboard loading and performance.
Create calculated fields where required.	Accurate and reliable data visualization.
Analyze crop production by state, year, season, and crop type.	Scalable to support future agricultural datasets.
Develop interactive dashboards with filters.	Easy to maintain and update.
Create Tableau Stories to present key findings.	Ensure smooth and responsive user experience.
Allow users to explore data through charts and maps.	Compatible with Tableau Desktop and Tableau Public.

3.3 Data Flow Diagram

The Data Flow Diagram (DFD) illustrates how data moves through the India's Agricultural Crop Production Analysis (1997–2021) project. The agricultural crop production dataset is collected from a reliable source and then cleaned, validated, and prepared for analysis. The processed data is used in Tableau to create interactive visualizations, dashboards, and stories. Finally, these dashboards present meaningful insights about crop production trends across different states, seasons, crops, and years, helping users analyze agricultural patterns effectively.



3.4 Technology Stack

Table-1 : Components & Technologies:

S.No.	Component	Description	Technology
1.	**User Interface**	Interactive dashboards for exploring crop production trends, yield patterns and state/crop-wise comparisons	Tableau Desktop / Tableau Public
2.	**Data Source**	India crop production dataset (1997-2021) with state, district,	CSV / Excel (data.gov.in / Kaggle - India Agriculture

S.No.	Component	Description	Technology
		crop, season, area, production and yield	Crop Production Dataset)
3.	**Data Cleaning & Preparation**	Handling missing values, standardizing state/crop/season names, correcting units and duplicate records	Tableau Prep Builder / MS Excel
4.	**Data Storage**	Storage of raw and cleaned dataset files used as the data source	Local Filesystem / Excel Workbook
5.	**Data Visualization Engine**	Building trend charts, maps, comparisons and calculated fields for year-wise, state-wise and crop-wise production analysis	Tableau Desktop
6.	**Dashboard Publishing**	Hosting and sharing interactive dashboards with end users	Tableau Public / Tableau Server
7.	**Report Generation**	Exporting summary insights and visuals for offline sharing	Tableau (Export as PDF / Image)
8.	**Infrastructure (Server / Cloud)**	Application accessed locally during development and published to the cloud for sharing	Local System, Tableau Public Cloud

Table-2: Application Characteristics:

S.No.	Characteristics	Description	Technology
1.	**Open-Source Frameworks**	List the open-source frameworks used	None (Tableau Public used as a freemium visualization tool)
2.	**Security Implementations**	Access controls implemented to protect the crop production dataset and published dashboards	Restricted local file access, Tableau Public/Server viewer permissions
3.	**Scalable Architecture**	Justify the scalability of the architecture (single Tableau workbook covering 25 years of data)	Tableau Data Extracts, incremental data refresh
4.	**Availability**	Justify the availability of the dashboard to end users	Tableau Public hosting with minimal downtime
5.	**Performance**	Design consideration for dashboard load speed with a large historical (1997-2021) dataset	Extracts (instead of live connection), filters, optimized calculated fields

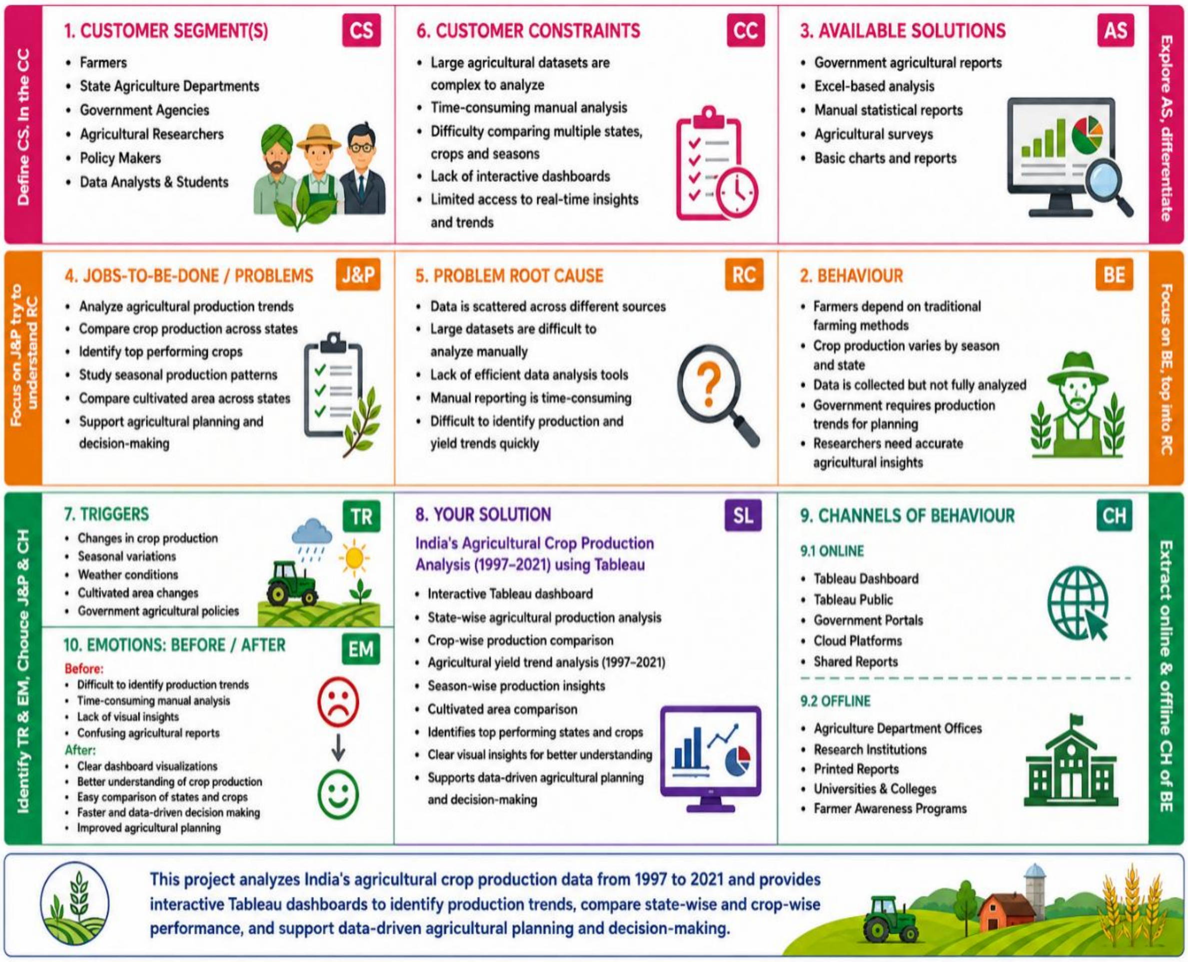
4. PROJECT DESIGN

4.1 Problem Solution Fit

Agricultural crop production data contains valuable information about crop yields, cultivated areas, states, and seasons. However, analyzing this information from raw data files is difficult because the data is large, complex, and not visually organized.

This project solves the problem by using Tableau to transform raw agricultural data into interactive visualizations. The dataset is first imported and cleaned to ensure accuracy. After preprocessing, different charts and dashboards are created to display crop production trends, state-wise comparisons, and yearly performance. Finally, a Tableau Story combines these visualizations to present the findings in a logical sequence.

The solution allows users to understand agricultural data quickly and make informed decisions based on visual insights instead of manually examining spreadsheets.



4.2 Proposed Solution

The proposed solution is an interactive agricultural data analysis system developed using Tableau. The project uses India's Agricultural Crop Production dataset (1997–2021) as the primary data source.

The workflow begins by importing the dataset into Tableau. The data is then cleaned by removing duplicate records, handling missing values, and verifying data types. Once the dataset is prepared, multiple visualizations such as bar charts, line charts, maps, and other graphical representations are created to analyze crop production across different states, years, seasons, and crop categories.

These visualizations are integrated into interactive dashboards with filters that allow users to explore the data dynamically. Finally, a Tableau Story is developed to present the analysis step by step and summarize the key findings. This approach improves data interpretation and supports effective agricultural analysis.

4.3 Solution Architecture

The solution architecture describes the overall workflow of the Agricultural Crop Production Analysis system. The project uses Tableau to transform raw agricultural data into meaningful visual insights through data preparation, analysis, and dashboard creation.

Solution Architecture Flow

1.Agricultural Dataset (1997–2021)

2.Data Preprocessing

- Remove duplicate records
- Handle missing values
- Verify data types
- Clean inconsistent values

3.Data Analysis in Tableau

- Import dataset into Tableau
- Create calculated fields (if required)
- Apply filters and sorting
- Analyze production, area, and yield trends

4.Dashboard Development

- KPI Cards (Total Production, Total Yield)

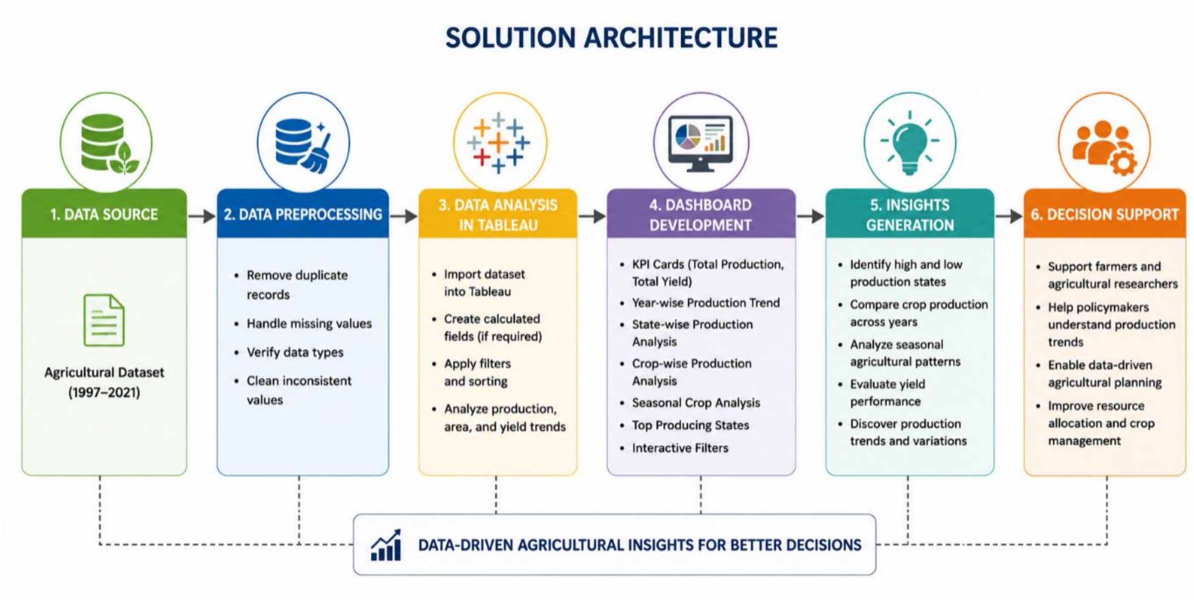
- Year-wise Production Trend
- State-wise Production Analysis
- Crop-wise Production Analysis
- Seasonal Crop Analysis
- Top Producing States
- Interactive Filters

5.Insights Generation

- Identify high and low production states
- Compare crop production across years
- Analyze seasonal agricultural patterns
- Evaluate yield performance

6.Decision Support

- Support farmers and agricultural researchers
- Help policymakers understand production trends
- Enable data-driven agricultural planning



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was planned in a systematic manner to ensure the successful completion of the agricultural crop production analysis using Tableau. The planning process included identifying the project objectives, collecting and preparing the dataset, performing data cleaning, designing visualizations, creating interactive dashboards, developing stories, and validating the final results.

The project was divided into the following phases:

- Requirement Analysis: Defined the project objectives and identified the required agricultural dataset.
- Data Collection: Gathered India's agricultural crop production data for the period 1997–2021.
- Data Preparation: Cleaned the dataset by removing duplicate records, handling missing values, and correcting data inconsistencies.
- Data Visualization: Created charts, graphs, and maps to analyze crop production across different states, districts, seasons, and crop types.
- Dashboard Development: Designed interactive dashboards with filters and parameters to provide meaningful insights.
- Story Creation: Combined multiple dashboards into a Tableau Story to present the analysis in a logical sequence.
- Testing and Validation: Verified the accuracy of calculations, visualizations, filters, and dashboard interactions.
- Documentation: Prepared the project report, presentation, and supporting documents for submission.

This structured project plan ensured that each phase was completed efficiently and contributed to the successful development of an interactive agricultural crop production analysis dashboard.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Performance testing was conducted to evaluate the efficiency, responsiveness, and reliability of the Tableau dashboard. The objective was to ensure that the dashboard performs smoothly while analyzing agricultural crop production data from 1997 to 2021.

The following performance aspects were tested:

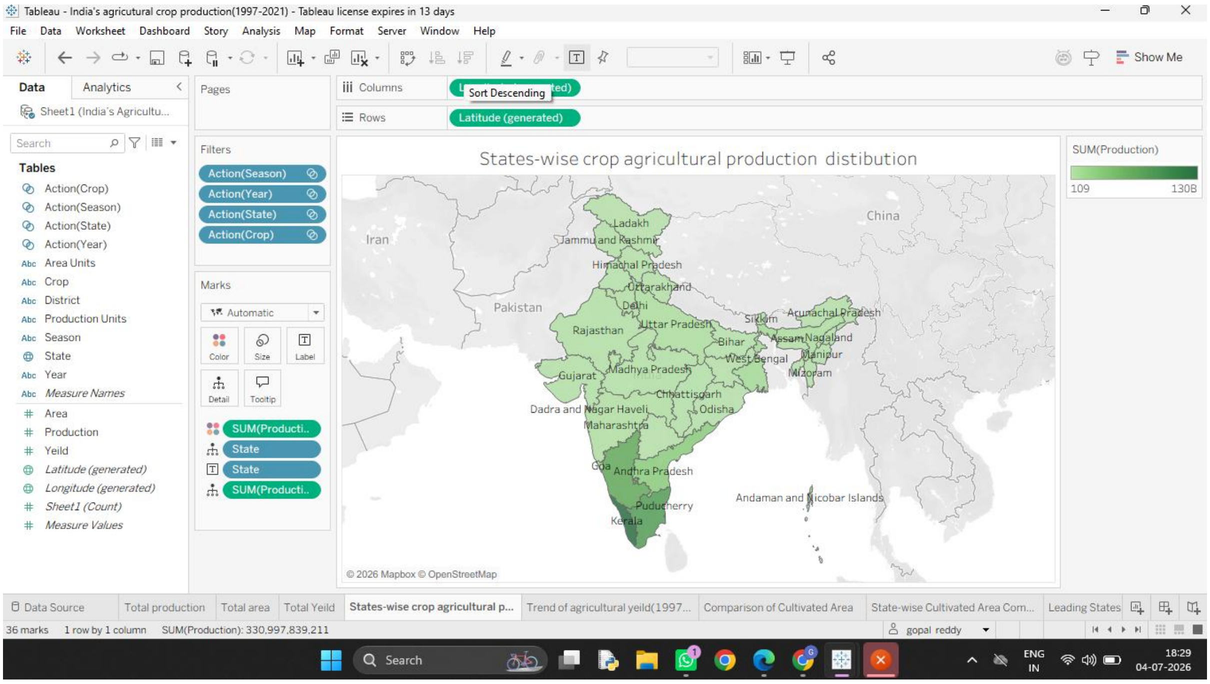
- Dashboard Loading: Verified that the dashboard loads within an acceptable time after opening the Tableau workbook or published dashboard.
- Filter Performance: Tested interactive filters such as State, District, Crop, Season, and Year to ensure they respond quickly and update the visualizations correctly.
- Chart Rendering: Confirmed that charts, maps, and graphs display accurately without delays or visual errors.
- Navigation Performance: Verified smooth navigation between worksheets, dashboards, and Tableau Stories.
- Calculation Accuracy: Checked calculated fields and aggregated values to ensure that the displayed results are accurate.
- Data Handling: Ensured that the dashboard efficiently processes the complete dataset without affecting performance.
- User Experience: Confirmed that the dashboard remains responsive and easy to use during interactive analysis.

The performance testing results showed that the Tableau dashboard functioned efficiently, responded quickly to user interactions, and provided accurate visualizations and insights without significant performance issues.

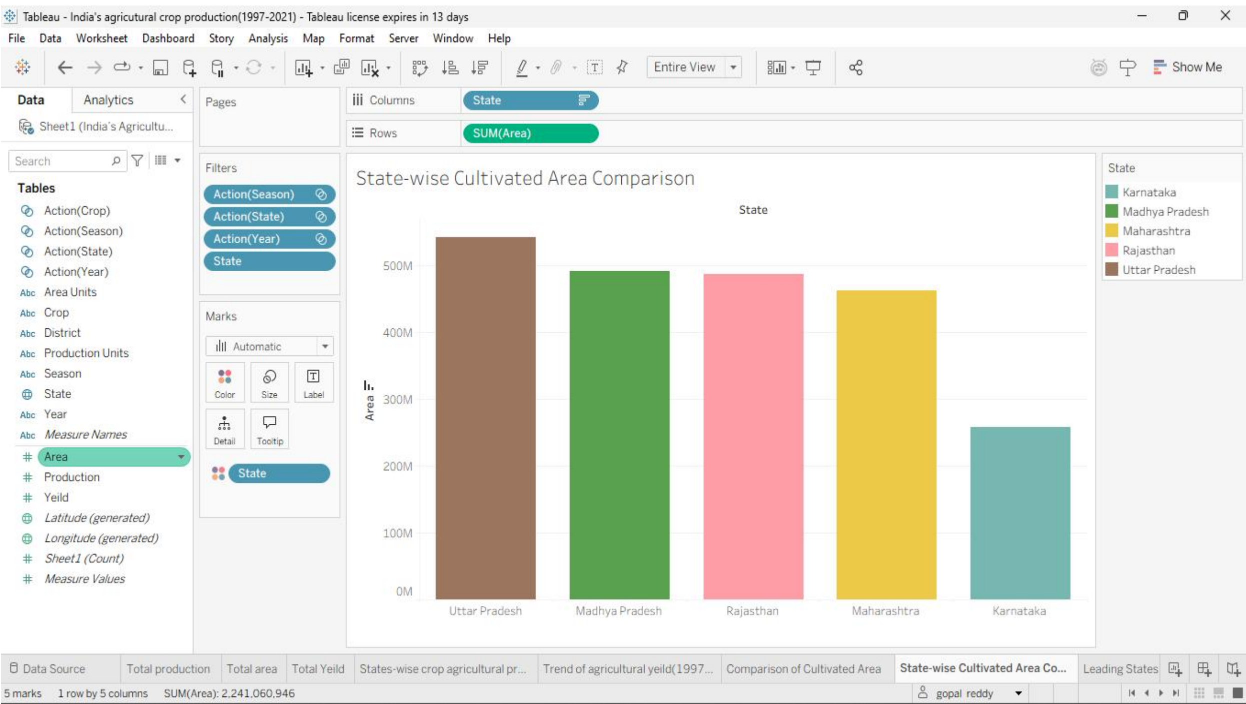
7. RESULTS

7.1 Output Screenshots

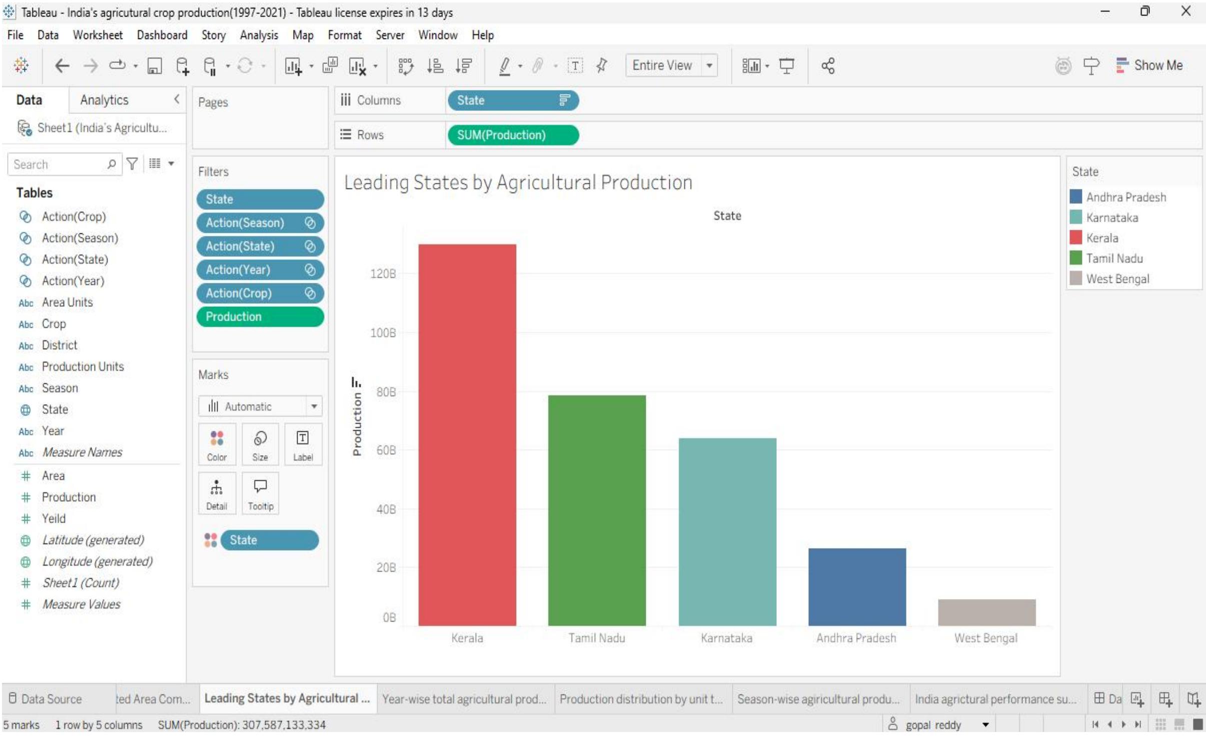
States-wise crop agricultural production distribution

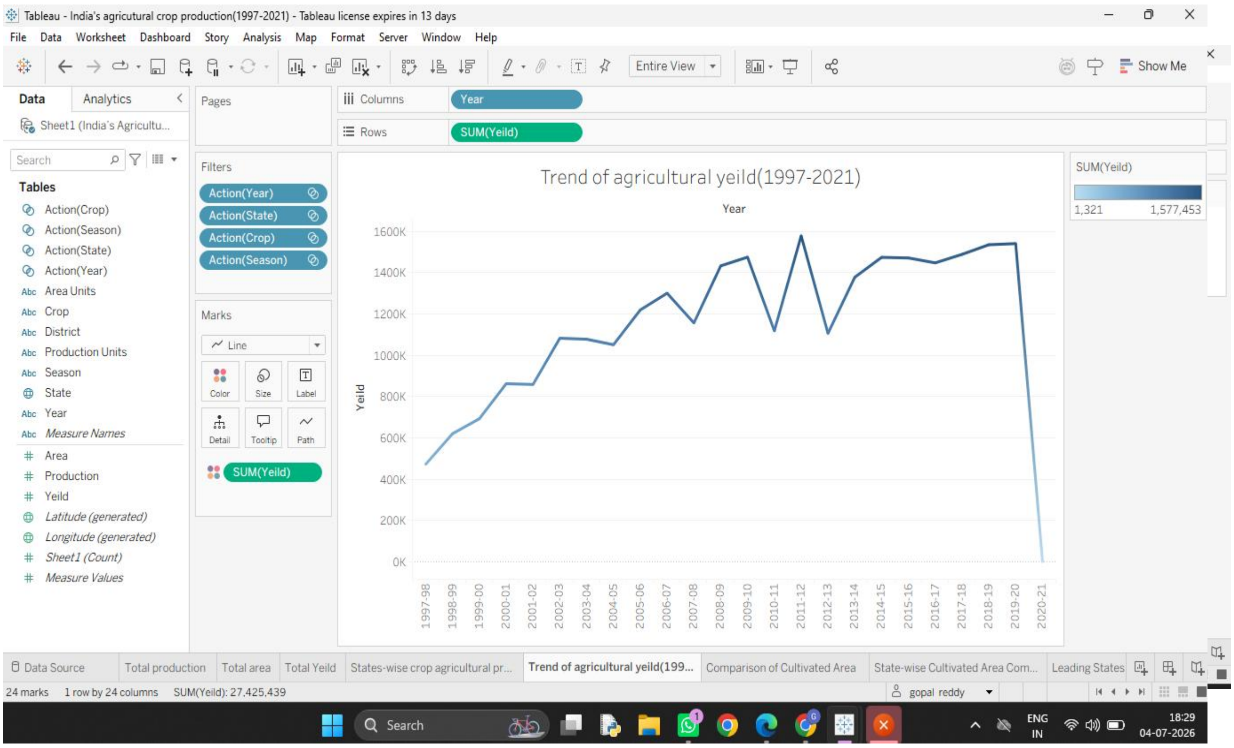


State-wise Cultivation area comparison



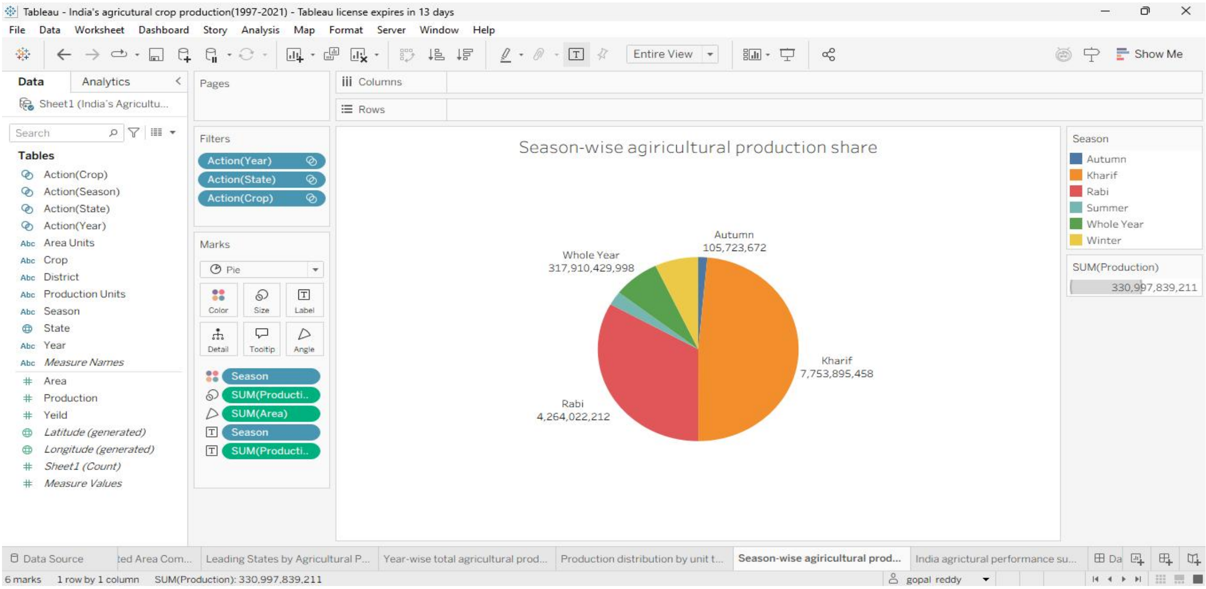
Leading States by Agricultural production



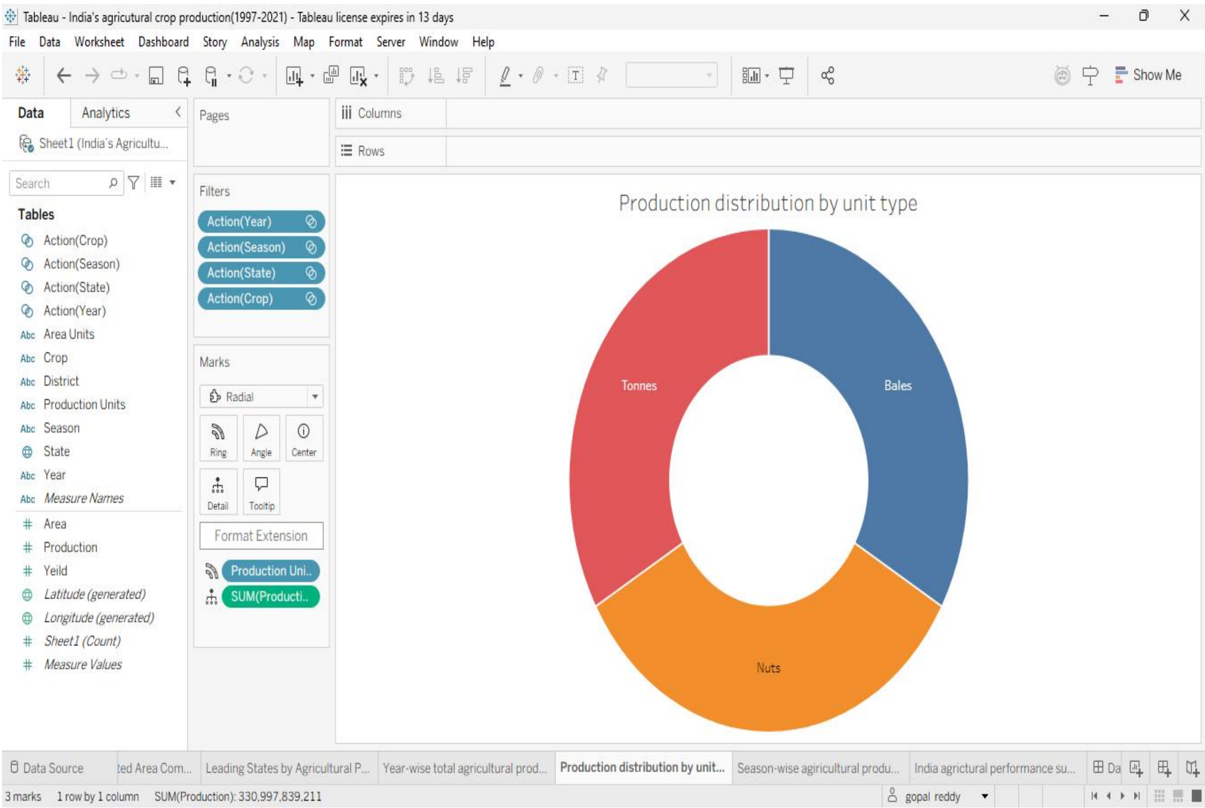


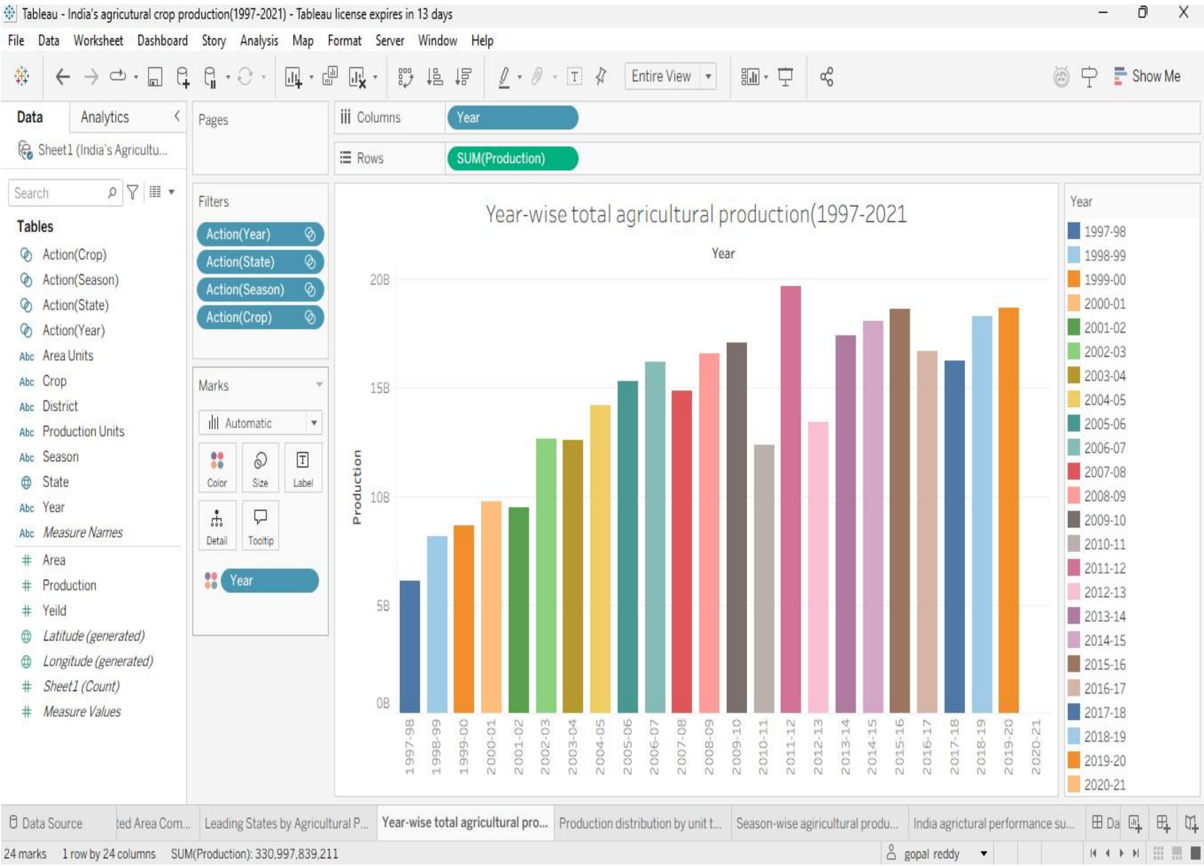
Trend of Agricultural Yeild(1997-2021)

Season-wise agricultural production share



Prodution distribution by unit type





Year-wise total agricultural production (1997-2021)

8. ADVANTAGES & DISADVANTAGES

Advantages

1. Interactive Visualization: Presents agricultural data through clear charts, graphs, and maps.
2. Easy Data Analysis: Enables quick analysis using filters for State, District, Crop, Season, and Year.
3. Supports Decision-Making: Helps users identify production trends and gain useful insights.

Disadvantages

1. Dependent on Data Quality: Results are accurate only if the dataset is complete and reliable.
2. Historical Analysis Only: The project analyzes data from 1997–2021 and does not predict future trends.
3. Manual Data Updates: New agricultural data must be added manually to keep the dashboard up to date.

9. CONCLUSION

The India's Agricultural Crop Production Analysis (1997–2021) project was successfully completed using Tableau. The project transformed raw agricultural data into interactive dashboards and visualizations, making it easier to analyze crop production across different states, crops, seasons, and years.

The developed dashboard helps users identify production trends, compare agricultural performance, and gain meaningful insights through interactive filters and charts. Overall, the project demonstrates how data visualization can support better understanding and informed decision-making in the agricultural sector.

10. FUTURE SCOPE

The project can be further enhanced by incorporating real-time agricultural data and expanding the analysis with additional factors such as rainfall, soil quality, and weather conditions. Predictive analytics and machine learning models can be integrated to forecast future crop production. The dashboard can also be deployed as a web-based application to provide easy access for farmers, researchers, and policymakers.

Links:

Data set link:

https://drive.google.com/file/d/1jgg2gFM7MZWG89I9RmB-qcClDYYEvdqH/view?usp=drivesdk

Github link:

https://github.com/gopalreddy-1810/India-s-Agricultural-Crop-Production-Analysis-1997-2021-

Dashboard and Story Links:

Dashboard-1:

https://public.tableau.com/app/profile/gopal.reddy1227/viz/Indiasagricuturalcropproduction1997-2021INCLUDESTORY/Dashboard1?publish=yes

Dashboard-2

https://public.tableau.com/app/profile/gopal.reddy1227/viz/Indiasagricuturalcropproduction1997-2021/Dashboard2?publish=yes

story:

https://public.tableau.com/app/profile/gopal.reddy1227/viz/Indiasagricuturalcropproduction1997-2021/Story1?publish=yes